CALCULUS OF INVERSE FUNCTIONS

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Name: Solutions

1. Find the limit: $\lim_{x\to\infty} 3^x$

$$\lim_{x\to\infty} 3^x = \boxed{\infty}$$

2. Find the limit: $\lim_{x\to-\infty} \left(\frac{1}{3}\right)^x$

$$\lim_{x\to-\infty} \left(\frac{1}{3}\right)^x = \lim_{x\to\infty} \left(\frac{1}{3}\right)^{-x} = \lim_{x\to\infty} 3^x = \boxed{\infty.}$$

3. Find $\frac{dy}{dx}$. Assume y is a differentiable function of x.

$$3y = xe^{5y}$$

$$\frac{d}{dx}(3y) = 3\frac{dy}{dx}$$

$$\frac{d}{dx}(xe^{5y}) = e^{5y} + x\left(5\frac{dy}{dx}\right)e^{5y}$$

$$3\frac{dy}{dx} = e^{5y} + 5xe^{5y}\frac{dy}{dx}$$

$$\Rightarrow e^{5y} = 3\frac{dy}{dx} - 5xe^{5y}\frac{dy}{dx} = \left(3 - 5xe^{5y}\right)\frac{dy}{dx}$$

$$\Rightarrow \boxed{\frac{dy}{dx} = \frac{e^{5y}}{3 - 5xe^{5y}}}$$

For problems 4-10, find $f'(x)$.

4. $f(x) = e^x \sin x$

$$f'(x) = \boxed{e^x \sin(x) + e^x \cos(x)}$$

5. $f(x) = \ln(xe^{7x})$

$$f'(x) = \frac{\frac{d}{dx}\left(xe^{7x}\right)}{xe^{7x}} = \frac{e^{7x} + x(7)e^{7x}}{xe^{7x}} = \frac{e^{7x}(1+7x)}{xe^{7x}} = \boxed{\frac{1+7x}{x}}$$

6. $f(x) = \dfrac{x}{\sqrt{1 - \ln(x)^2}}$

$$f'(x) = \frac{\sqrt{1-\ln(x)^2} - x\left(\frac{1}{2}\right)\left(1-\ln(x)^2\right)^{-1/2}\left(\frac{-2\ln(x)}{x}\right)}{\left(1-\ln(x)^2\right)}$$

$$= \boxed{\frac{\sqrt{1-\ln(x)^2} + \frac{\ln(x)}{\sqrt{1-\ln(x)^2}}}{\left(1-\ln(x)^2\right)}}$$

7. $f(x) = (xe^x)^\pi$

$$f'(x) = \pi(xe^x)^{\pi-1}\frac{d}{dx}(xe^x)$$

$$= \boxed{\pi(xe^x)^{\pi-1}(e^x + xe^x)}$$

8. $f(x) = (e^{2x} + e)^{\frac{1}{2}}$

$$f'(x) = \frac{1}{2}(e^{2x}+e)^{-1/2}(2e^{2x})$$

$$= \boxed{\frac{e^{2x}}{\sqrt{e^{2x}+e}}}$$

9. $f(x) = \Big(\ln(5x^2 + 9)\Big)^3$

$$f'(x) = 3(\ln(5x^2+9))^2\frac{d}{dx}\ln(5x^2+9)$$

$$= 3(\ln(5x^2+9))^2\frac{\frac{d}{dx}(5x^2+9)}{5x^2+9}$$

$$= \frac{3(\ln(5x^2+9))^2(10x)}{5x^2+9}$$

$$= \boxed{\frac{30x(\ln(5x^2+9))^2}{5x^2+9}}$$

10. $f(x) = \ln\left((5x^2+9)^3\right)$

$$f'(x) = \frac{\frac{d}{dx}\left((5x^2+9)^3\right)}{(5x^2+9)^3} = \frac{3(5x^2+9)^2(10x)}{(5x^2+9)^3}$$

$$= \boxed{\frac{30x}{5x^2+9}}$$

For problems 11-20, find the indefinite integral (you may need u-substitution).

11. $\int e^x \, dx = e^x + C.$

12. $\int a^x \, dx$, where $a > 0$ is a constant ($\neq 1$).

$a^x = e^{\ln(a)x}$ $\qquad u = \ln(a)x \Rightarrow \frac{du}{\ln(a)} = dx$

$$\int a^x dx = \int e^{\ln(a)x} dx = \frac{1}{\ln(a)} \int e^u du = \frac{1}{\ln(a)} e^u + C = \boxed{\frac{1}{\ln(a)} a^x + C}$$

13. $\int \pi^{2x} \, dx$

$u = 2x \Rightarrow du = 2dx$

$\Rightarrow \frac{1}{2}du = dx$

$$\int \pi^{2x} dx = \frac{1}{2}\int \pi^u du = \frac{1}{2}\left(\frac{1}{\ln(\pi)} \pi^u\right) + C$$

$$= \boxed{\frac{1}{2\ln(\pi)} \pi^{2x} + C}$$

14. $\int \frac{1}{x}\,\mathrm{d}x = \boxed{\ln|x| + C}$

15. $\int e^{2x}\,\mathrm{d}x$

$u = 2x \Rightarrow du = 2dx \Rightarrow \frac{1}{2}du = dx$

$\int e^{2x}dx = \frac{1}{2}\int e^{u}du = \frac{1}{2}e^{u} + C = \boxed{\frac{1}{2}e^{2x} + C}$

16. $\int \frac{\ln x}{x}\,\mathrm{d}x$

$u = \ln(x) \Rightarrow du = \frac{1}{x}dx$

$\int \frac{\ln(x)}{x}dx = \int u\,du = \frac{1}{2}u^2 + C = \boxed{\frac{1}{2}\ln(x)^2 + C}$

17. $\int \frac{\sqrt{\ln(x)}}{x}\,\mathrm{d}x$

$u = \ln(x) \Rightarrow du = \frac{1}{x}dx$

$\int \frac{\sqrt{\ln(x)}}{x}dx = \int \sqrt{u}\,du = \frac{2}{3}u^{3/2} + C = \boxed{\frac{2}{3}\ln(x)^{3/2} + C}$

18. $\int \frac{e^x}{\sqrt{1-e^x}}\,\mathrm{d}x$ $\quad u = 1-e^x \Rightarrow du = e^x dx$

$$\int \frac{e^x}{\sqrt{1-e^x}}dx = \int \frac{1}{\sqrt{u}}du = \int u^{-1/2}du = 2u^{1/2}+C$$

$$= \boxed{2\sqrt{1-e^x}+C}$$

19. $\int \frac{\ln(e^{2x})}{x^2}\,\mathrm{d}x$ $\quad u = x^2 \Rightarrow du = 2x dx$

$$\int \frac{\ln(e^{2x})}{x^2}dx = \int \frac{2x}{x^2}dx = \int \frac{1}{u}du = \ln(u)+C = \boxed{\ln(x^2)+C}$$

$$= \boxed{2\ln(x)+C}$$

20. $\int \frac{e^x}{3+e^x}\,\mathrm{d}x$ $\quad u = 3+e^x \Rightarrow du = e^x dx$

$$\int \frac{e^x}{3+e^x}dx = \int \frac{1}{u}du = \ln(u)+C = \boxed{\ln(3+e^x)+C}$$

21. Evaluate the definite integral: $\int_2^3 \frac{xe^{x^2}}{3}\,\mathrm{d}x$ $\quad u = x^2 \Rightarrow du = 2x dx \Rightarrow \frac{1}{2}du = x dx$

$$\int_2^3 \frac{xe^{x^2}}{3}dx = \frac{1}{6}\int_4^9 e^u du = \boxed{\frac{1}{6}(e^9-e^4)}$$